Mandeep Khatri

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EDUCATION

Alabama A&M University, *Huntsville, AL* ***Expected Graduation: May 2028***

Bachelor of Science in Computer Science | Minor: Finance *GPA: 3.85/4.0*

Awards & Honors: Dean's List, AAMU Merit Full Scholarship, EICOP 2025 Finalist

Relevant Coursework: Data Structures & Algorithms, Intro to programming I and II (Python), Computer Architecture, Software Engineering, Operating Systems, Discrete Mathematics, Linear Algebra

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript, TypeScript, SQL, MATLAB, Go

Frameworks & Libraries: React, React Native, NestJS, Flask, pandas, NumPy, Streamlit, Socket.IO, LangChain, FAISS

Cloud & Tools: AWS (EC2, Lambda), GCP (Vertex AI), Docker, PostgreSQL, Prisma, Git, GitLab CI, CUDA, Linux

WORK EXPERIENCE

Research Engineer - Autonomous System Lab, *Alabama A&M University, Huntsville, AL* ***October 2025 – Present***

- Led a rewrite of the drone perception pipeline (**Canny + Lucas–Kanade**) from **MATLAB** to **C++/CUDA**; restructured memory access, cutting per-frame latency from 84ms to 27ms on Jetson Orin (30 FPS).
- Drove the lab's first automated perception testing: built a **Python-based** Gazebo regression harness over 14 scenarios, gated every merge via **GitLab CI**, and caught a frame-skip bug before flight test.
- Proposed and built a Kalman-filter sensor-fusion module in **C++** that smooths IMU + camera streams in the perception stack, cutting simulated trajectory jitter 38%.

Teaching Assistant, *Alabama A&M University, Huntsville, AL* **January 2025 – Present**

- Built a **pytest autograder** for CS104 (Python) and CS109 (C++), adopted by the TA team across 35-student lab sections; cut per-assignment grading from 2.5h to 25min and rewrote 4 lab handouts after student feedback.
- Started a weekly LeetCode workshop for freshmen prepping for internships; ran a 12-week DSA track (arrays, hash maps, graphs, DP) on GitHub Classroom with mock-interview sessions, growing to 15+ regulars.

Research Assistant, *Alabama A&M University, Huntsville, AL* ***March 2024 – December 2025***

- Wrote a MATLAB + Python pipeline ingesting 1,100+ satellite-tracked ocean drifter trajectories from NOAA NetCDF (anomaly filtering, velocity-magnitude, lat/lon resampling); spatial convergence modeling recovered two subtropical gyre centers, matching published South Pacific coordinates within ~0.4°.
- Set up a reproducible Python workflow (Matplotlib + cartopy) for trajectory and velocity visualizations, version-controlled in the lab's Git repo; two other students reused it for follow-on datasets.

Software Engineering Intern, *Verisk, Remote* ***August 2023 – December 2023***

- Added new sensor types in the team's IoT telemetry service (Python, MQTT, AWS EC2); shipped a JavaScript filter on the Chart.js dashboard, a Flask CSV endpoint, 47 PyTest cases, & helped Dockerize the service.
- Investigated slow REST API responses under JMeter load testing (~300 RPS) and worked with my mentor to identify a Postgres connection pool issue; after the fix, p95 API latency dropped from ~1.4s to ~820ms.

PROJECTS

Route Intelligence - *Food Rescue Platform* ***JPMorgan Chase Hackathon, 2026***

- Processed 150K+ rescue records (6.67M lbs of food) using Python and Pandas across 4 cities; a ZIP-3 batching pass flagged 845 same-day pickup pairs, unlocking $106K in potential logistics savings.
- Built a 4-tier risk score (recency, claim rate, pickup size) on a Folium + Streamlit map; 35% of rescues fell into Watch/Elevated/High.

Internably – *Student Networking Platform* ***Present***

- Built an iOS app for college internship networking with a React Native + TypeScript client & NestJS + Postgres + Prisma backend; wrote the auth flow with .edu gating, JWT refresh-token rotation, and Socket.IO for DMs.
- Set up an npm workspaces monorepo with three shared packages (UI tokens, TS types, ESLint config) for the mobile and planned web client; TanStack Query for caching, Zustand for state.

LLM Course Q&A Chatbot ***October 2025***

- Built a RAG pipeline ingesting 100+ pages of course PDFs using PyMuPDF, recursive text splitting, and Gemini embeddings via Vertex AI, achieving 85% manual Q&A accuracy against a ground-truth evaluation set.
- Optimized FAISS dense retrieval and prompt engineering templates, reducing average query response latency by 40% compared to a keyword search baseline, improving real-time usability for students.

ACTIVITIES

- **NRF Foundation Ray Greenly Scholar** - *Competitive National Selection* **January 2026**
- **Google Cloud Career Jumpstart** - *HBCU Fellow* **May 2025 – August 2025**
- **Propel2Excel Fellow** - *selective mentorship cohort* **September 2025 – Present**
- **Innovate Alabama AI & Public Safety Hackathon** - *Competitor* **November 2025**
- **TracerFire Cybersecurity Competition** - *Competitor, Sandia National Laboratories* **September 2024**